Controlled cue learning in a non-renewing body

Externally arranged encounters; fast delta and eligibility both succeed. No new-method advantage.

A One continuous conservative body; no tissue birth

ACQUIRE
cue A + / cue B −

2 epochs; 8 outcomes per cue

REVERSE
cue A − / cue B +

After step 512; not signalled

NEUTRAL
64 neutral steps

Then an unrewarded probe

RETAIN / ERASE
+64-step probes

Only Q reset to 0.2

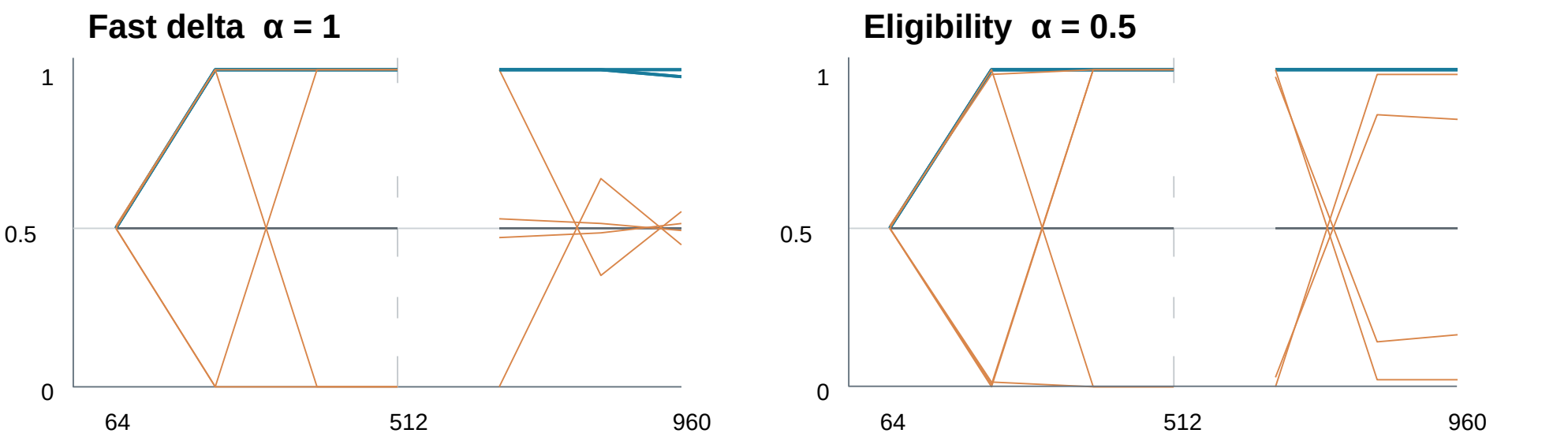
Mapping shown for order 0; order 1 is reversed. Initial probe: 1–64. A neutral gap also follows acquisition.

Local cue registration → delayed local outcome → Q update → shared cue-gradient actuator

Same transported tissue, energy and sensor registrations throughout; no reset at reversal.

B Both rules learn; unpaired feedback yields opposing individual preferences

Correct-cue contact mass fraction • 2 seeds × 2 mappings per condition • saved probe endpoints

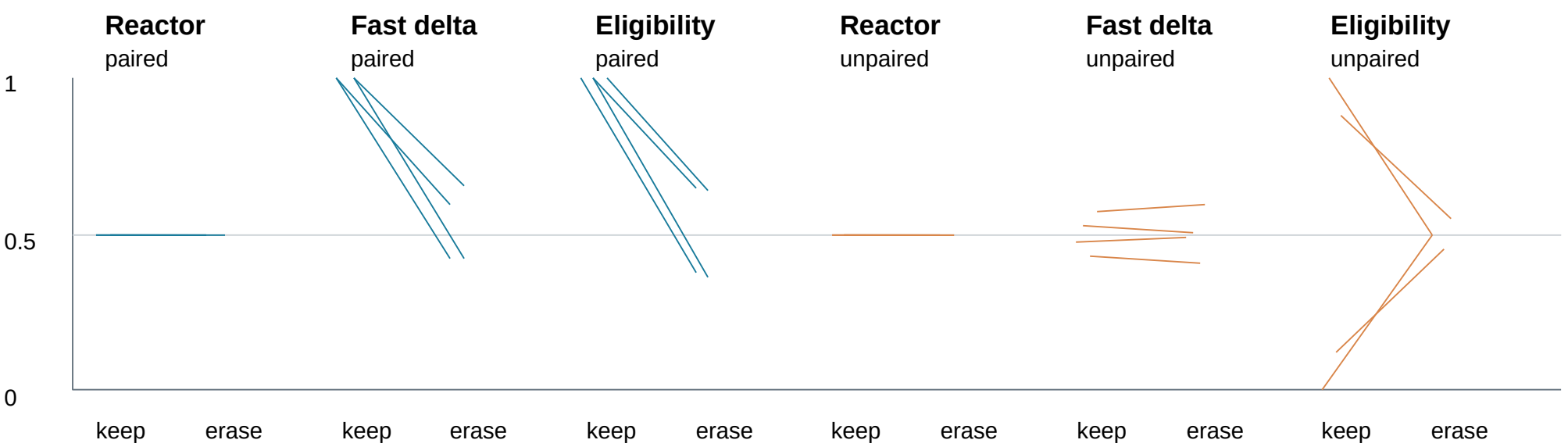


Step 512: relation switches. First reversal probe follows 8 feedback pulses; immediate recovery was not measured.

Paired Unpaired Reactor (no Q writes)

C The same terminal body: preserve Q or erase Q, with all feedback off

Correct-cue contact fraction at +64 steps; writes off in both forks. Body, E, last-cue and trace are retained.



2 seeds × 2 mappings = 4 dependent conditions per arm/pairing; 24 lives + 48 forks. No cell-level n.

All survive; M ≈ 24, no births. Life E: 72 → 38.66–42.29 EU. Basal + motion paid; Q writes not separately priced.

Two installed local plasticity rules and a shared actuator; these results do not establish new-method superiority.